



Threat Modeling of ML-intensive Systems: Research Proposal

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ABSTRACT

Context: The rise of Artificial Intelligence (AI) and Machine Learning (ML) applied in many software-intensive products and services introduces new opportunities but also new security challenges. **Motivation:** AI and ML will gain even more attention from industry in the future, but threats caused by already discovered attacks specifically targeting ML models are either overseen, ignored, or mishandled. **Problem Statement:** Current Software Engineering security practices and tools are insufficient to detect and mitigate ML Threats systematically. **Contribution:** We will develop and evaluate a threat modeling technique for non-security experts assessing ML-intensive systems in close collaboration with industry and academia.

KEYWORDS

Adversarial Machine Learning, Threat Modeling, Industry

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1 INTRODUCTION

Software Engineering (SE) and IT Security are two established research areas contributing to practices that address security issues. Artificial Intelligence (AI) and its subfields—Machine (ML) and Deep-learning (DL)—are being implemented in products and services in critical sectors, such as medicine, telecommunication, and energy.¹ There are only a small number of security regulations in place, most of which are non-technical with little to no guidance for engineers (e.g., the European Union AI Act²). Without guidance for practitioners to secure these systems, malicious actors have a plethora of opportunities for exploiting unaddressed vulnerabilities.

The number of known AI Incidents and Controversies reported by the AI, Algorithmic, and Automation Incidents and Controversies (AIAAIC) Repository increased from 34 cases in 2021 to 76 reported in 2022.³ Such incidents are likely to increase given the lack of technical guidance in tackling new security issues related

to ML. Systems relying on ML models—i.e., ML-intensive systems [1]—are subject to attacks; therefore, security assessments, such as threat modeling, need to be in place to identify and document potential threats that can exploit vulnerabilities and mitigate them [2]. In this context, the set of attacking and defending ML models is called Adversarial Machine Learning [7]. This thesis will investigate threat modeling for ML-intensive systems and provide mitigation suggestions. In particular, the objective of my thesis will be to develop a comprehensive threat modeling technique assisting software engineers in identifying and documenting potential security threats associated with a software product. The proposed threat modeling technique will target software engineers without previous expertise assessing ML-intensive software systems. It will be partly based on available threat modeling techniques that can be transferred and used effectively in the development of ML-intensive systems in close collaboration with industry. To mitigate the identified threats, I propose to start by applying non-ML mitigation techniques.

My contribution will be the development and iterative evaluation of the proposed threat modeling technique with relevant industries. The threat modeling technique will ease the process of identifying potential attack vectors threatening ML-intensive systems. Thus, we need a method built upon common practices to reveal potential attacks.

Problem Statement. Current security assessment techniques are barely researched and insufficient to address the unique challenges inherent to engineering ML-intensive software systems. To rectify this, it is crucial for practitioners to be aware of the potential ML threats, comprehensively identify them, facilitate the development of ML-specific threat models, and suggest both ML and non-ML mitigations and security practices.

2 BACKGROUND

Adversarial Machine Learning (AML) investigates malicious third-party actions targeting ML models to force them to either misclassify [4], reduce accuracy [14], and steal training data [3, 12]. These AML attacks are theoretically feasible [13] and practically transferable from the laboratory to real-world scenarios [8]. As a matter of fact, practitioners already have experienced AML [6] and reported a shortage of tools to identify and mitigate ML threats [10].

My ongoing literature study shows a lack of collaboration between academia and industry, confirming the observations by the secondary study of Apruzzese et al. [1]. The authors further suggest adapting and applying threat modeling techniques to realistic ML-intensive software systems in real-world scenarios instead of synthetic toy examples created in a laboratory environment. AML mitigation techniques proposed in the literature, such as adversarial training to defend against Adversarial Examples, are impractical, while non-ML mitigation practices are considered more beneficial

¹<https://aiindex.stanford.edu/report/>

²<https://www.europarl.europa.eu/news/en/headlines/society/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence>

³<https://www.aiaaic.org/aiaaic-repository/ai-algorithmic-and-automation-incidents>

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[11]. So far, threat modeling techniques for ML-intensive systems utilize Dataflow Diagrams and Microsoft STRIDE [16] or an asset-centered approach of STRIDE [9] to identify ML threats. Other modeling technique focus on attacker strategies extracted from experience reports [15], Cost-driven threat approaches [1], and the probability of exploitation of an ML threat [11]. Although all studies utilize common industry practices and a cost-driven approach to threat modeling is compelling for practitioners, developing an ML threat modeling technique with practitioners has not been the focus of researchers.

3 RESEARCH HYPOTHESIS AND EXPECTED CONTRIBUTIONS

Based on our ongoing studies and industry collaborations, I hypothesize that an ML threat modeling technique will support software engineers without AML expertise in securing ML-intensive systems from AML attacks. To achieve this, I will investigate industrial practices and needs and analyze the propositions in the literature regarding threat modeling for ML-intensive systems. I will adapt ML threat modeling techniques to accommodate those needs.

As a first step, I need to identify all ML threat modeling techniques proposed in the literature to compare the existing work and extract useful bits contributing to assembling a comprehensive threat modeling technique. This leads us to the following research question (RQ):

- **RQ1** What is the state-of-the-art reported in the literature of ML Threat Modeling of ML-intensive Software Systems?

The results will provide an overview of ML threat modeling techniques, including their challenges and benefits, which will serve as a foundation to compare them to current ML threat modeling used in practice. This brings me to the next research questions, which we aim to address in collaborations with our industry partners:

- **RQ2** What challenges and needs do practitioners face when assessing the security of ML-intensive Software Systems?

To answer RQ2, I will conduct a qualitative survey where we plan to interview practitioners about their experience when they assessed ML threats of ML-intensive systems by applying common threat modeling techniques. Since numerous challenges and needs will probably emerge while conducting the survey, I will formulate interview questions addressing the following aspects: 1) the accuracy of those methods with regard to ML threat detection accuracy, 2) the feasibility of the current threat modeling techniques applied in practice for non-AML experts and 3) their suitability to assess design decisions to integrate an ML model into an existing system, turning it into an ML-intensive Software System. After collecting all the necessary information, I will compare current and discovered new or common challenges and benefits of threat modeling techniques discussed in academia and industry. These insights provide a structured overview to investigate which existing SE threat modeling techniques can address most of the challenges during the engineering of ML-intensive Software Systems, resulting in the next RQ:

- **RQ3** How and to what extent can we adapt traditional Threat Modeling Techniques to the engineering of ML-intensive Software Systems?

After determining suitable ML threat modeling techniques, we assemble the first version of a threat modeling technique that is suitable for non-AML experts to assess ML-intensive Software Systems. I need to evaluate its performance, leading us to the last RQ:

- **RQ4** What are the benefits and limitations of adapting Threat Modeling Techniques to ML-intensive Software Systems?

RQ3 and RQ4 encompass our evaluation process, where I will iteratively involve practitioners to evaluate the proposed threat modeling technique. Depending on the practitioners' feedback, I will re-iterate through existing threat modeling techniques to determine if they solve the issues reported by the practitioners or if an alternative solution is required. This thesis, in particular the development and iterative evaluation (RQ3 and 4), follows the steps proposed in the technology transfer model [5] in analogy to design science research for software engineering. My research will deliver an ML threat modeling technique, which will provide an overview of ML Threats for a given system and support the decision on which mitigation practices to apply.

4 CONCLUSION

The introduction of ML into a software system is a new security challenge that needs to be jointly solved by SE and IT Security practitioners and researchers. My research aims to create an ML threat modeling technique in close collaboration with practitioners.

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